

Data Question

Who would win in a chess match? OpenAI's ChatGPT or Anthropic's Claude AI? That is the question I came to for this final project. And, as it turns out, Claude AI was the better of the two AIs. What inspired me to reach this question was earlier in this semester, while listening intently to a professor's lecture during another class on Beetlejuice 5 (I believe) I had started up a prompt for the AI called ChatGPT and ended up playing some rousing a few games of tic tac toe against the AI. While playing those games I noticed two things: one being it seemed the AI was letting me win each game, and two after calling it out on that fact it began to cheat after a what seemed a set number of terms or games. That got me thinking: can ChatGPT play chess, and can I make ChatGPT play against another AI? Once I found that I could the question became: Which AI would win the chess game?

Data Provenance and Collection Methods

To reach the answer to the basis of this project, I created two programs. The first program was designed to connect two AIs via API calls, create a chess board game environment to connect the two AI's long enough to play a few games of chess, and to collect all datapoints I found to be useful to find the answer to my question. The second program was designed to clean the collected raw data to then parse through it to get easy to read statistics for analysis.

The Two AI's I decided to work with was OpenAI model, gpt-4-turbo and Anthropic Claude-sonnet-4-5-20250929. The latest and comparatively intelligent Large Language Model (LLM) AIs. To communicate with the two AIs I created a system prompt to both AIs that stated:

"You are a chess engine playing as White. Your response must begin with a single, valid, and legal UCI chess move for White on the current board (e.g., "e2e4"). Optionally, you may follow the move with a brief comment about your move on a new line, starting with "Comment: ". Example output: "e2e4Comment: I'm deploying my king's pawn." DO NOT share your strategy."

This same message was repeated to the AI that was playing as the Black Chess pieces. Although I initially believed this would be easy an easy project to prompt a statement to play chess, then gather needed data, it turns out that I needed to edit this prompt at least 25 times to reach the above message prompt. Another challenge I came across was using Regular Expressions for the first time in a long while to remove any unneeded communications between the AIs to only collect data that I felt I needed. It was what probably made this project harder than it would have been, but I believe the payoff was worth it. The other challenge that I faced was interpreting into code the two different rate limit methodologies used by both OpenAI and Anthropic.

Dataset Details and Organization

The strategy was simple. Play have the two AIs play 150 games of chess, 75 would be games where the AIs starting positions (what color pieces they each played) would be fixed. The next 75 games were designed to allow the winner of a game play as the white pieces (make the first move) for the next game. I also had 35 datapoints that were to be collected from each game and managed to generate 815 datapoints in total from all 150 games. I had also added a system that

would not allow any illegal chess moves with a three-strike warning to the AI. This part of the system however became a bit of a problem later since these AIs are LLM's, they are "talkative" meaning to conduct an action, they will describe it back to you or make some sort of comment. Which is why the prompt and Regular Expressions becoming so important, and the bane of my existence for a time, besides the AI rate limits. As a result, the data I was able to capture and clean was interesting to say the least.

To generate any useful information from the games I saved the games and their move data over to a json file that I was then able to flatten and turn into a CSV file that I could then use the python library Panda to sort and present the data I found. In total, gpt-4-turbo won 22 of the 150 games, where Claude-Sonnet-4-5 won 78 out of 150 games. What was also found was during each game there was an instance of one of the AIs attempting to use an illegal move. The total illegal moves by AI during the control games were 210 white, and 192 for black. This told me that gpt-4-5-turbo was "cheating" to win or at the very least got confused during each match. There was also a pattern to use the same white paw (e2) during each game.

Analysis and Findings

With an overall result, Claude-sonnet-4-5 was the better chess player when compared to gpt-4-5-turbo. I did try to create visuals for my data, but I had messed up the use of my time management and could not unfortunately provide.

My analytical approach to this project was to break the project down into stages. First research the API requirements for rate limits, and prompt format for the two APIs. The second was to create or find an existing chess program that would allow me to connect two AI's using API call to communicate. However, at the end of the day I needed to make my own chess board for the AIs. The next was to determine how to word a prompt that would express my needs to the AIs.

What I did find outside of the statistical data, was that the AIs needed to "reprimanded" or be made aware that they were making illegal moves and were being punished for them by telling them they had lost because of x illegal moves. After a few games it needed the AIs began to have longer chess games and even use more than the 4 pans in to control the center of the boar. That was another interesting point in this research I found. Both AIs regardless of color they were playing always attempted to control the center of the board each game. When that happened, one of both AIs during that game would then start moving the White pieces in ways that would be considered illegal.

Ethical Considerations

There were two points regarding ethics that I felt needed to be considered for this type of data. Responsibility: the users of these programs needed to keep in mind the terms of service for both OpenAI API and Anthropic's API. Another point of responsibility is to not use the data to mislead others with the data gathered from this project.